Complex Numbers

Section 1: Complex Arithmetic & Polar Form

1 What is the polar (trigonometric) form of the complex number $z = -2 + 2i\sqrt{3}$?

a. $4(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})$

b. $4(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$

c. $2(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})$

d. $4(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6})$

2 What is the product of $z_1 = 3(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$ and $z_2 = 2(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})$?

a. $6(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4})$

b. $5(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4})$

c. $6(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$

d. $6(\cos\frac{\pi}{8} + i\sin\frac{\pi}{8})$

3 Simplify the quotient $\frac{8(\cos 80^\circ + i\sin 80^\circ)}{2(\cos 20^\circ + i\sin 20^\circ)}$.

a. $4(\cos 60^\circ + i\sin 60^\circ)$

b. $4(\cos 100^\circ + i\sin 100^\circ)$

c. $4(\cos 4^\circ + i\sin 4^\circ)$

d. $6(\cos 60^\circ + i\sin 60^\circ)$

Section 2: De Moivre's Theorem & Powers

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4 Using De Moivre's Theorem, evaluate $(1+i)^8$.

a. 16

b. $16i$

c. 32

d. -16

5 What is $(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6})^{12}$?

a. 1

b. -1

c. i

d. $-i$

6 What is the real part of the complex number $(2-2i)^4$?

a. -64

b. 64

c. 0

d. -32

Section 3: Roots of Complex Numbers & Roots of Unity

7 How many distinct complex solutions exist for the equation $z^5=1$?

a. 1

b. 3

c. 5

d. Infinitely many

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8 Which of the following is a primitive cube root of unity?

a. $-\frac{1}{2} + i\frac{\sqrt{3}}{2}$

b. $\frac{1}{2} + i\frac{\sqrt{3}}{2}$

c. $-\frac{1}{2} - i\frac{1}{2}$

d. $1 + i\sqrt{3}$

9 What is the sum of all the n-th roots of unity for any integer $n \geq 2$?

a. 0

b. 1

c. n

d. -1

Section 4: Complex Polynomials & Geometry

10 If $2 + 3i$ is a root of the quadratic polynomial $x^2 + bx + c = 0$ with real coefficients, what is the value of $b + c$?

a. 9

b. 11

c. 13

d. 17

11 In the complex plane, what geometric figure is described by the equation $|z - (3 + 4i)| = 5$?

a. A circle centered at $(3, 4)$ with radius 5

b. An ellipse with foci at $(3, 4)$

c. A line passing through $(3, 4)$

d. A parabola with vertex $(3, 4)$

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12 What is the minimum value of $|z - 6|$ given that $|z| = 2$?

a. 2

b. 4

c. 6

d. 8

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